**4** Fig. 4.1 shows a heater used to warm the air in a room.

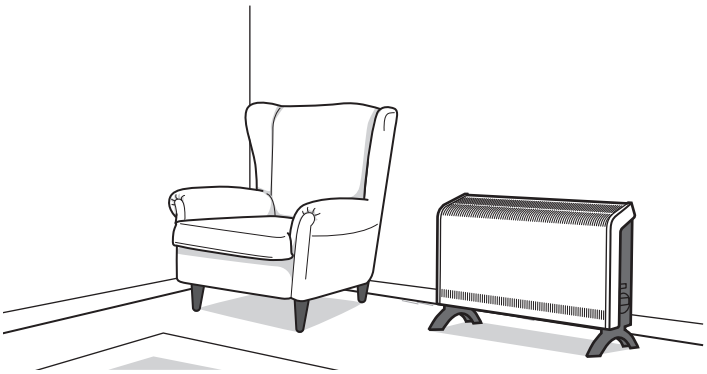

**Fig. 4.1**

**(a) (i)** State the main method of thermal energy transfer throughout the air in the room.

..... [1]

**(ii)** Explain how the heater warms **all** the air in the room.

.....

.....

.....

.....

..... [3]

**(b)** The power of the heater is 2.0 kW when it is connected to the mains supply with an e.m.f. of 230 V.

**(i)** Show that the current in the heater is approximately 8.7 A.

[2]

**(ii)** The plug connecting the heater to the mains supply is fitted with a fuse.

Fuse ratings of 3A, 5A, 10A and 13A are available.

State which fuse is used. Explain your answer.

fuse .....

explanation .....

.....

..... [2]

[Total: 8]

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